Kubernetes Components and Helm Chart Assignment

1. Kubernetes Installation Steps

The following diagram shows a simplified Kubernetes installation workflow:

```
+-------------------+
| Prepare Servers   |
| (Linux Machines)  |
+---------+---------+
          |
          v
+-------------------+
| Install Container |
| Runtime           |
| (containerd)      |
+---------+---------+
          |
          v
+-------------------+
| Install kubeadm,  |
| kubelet, kubectl  |
+---------+---------+
          |
          v
+-------------------+
| Initialize Master |
| kubeadm init      |
+---------+---------+
          |
          v
+-------------------+
| Configure kubectl |
+---------+---------+
          |
          v
+-------------------+
| Install CNI       |
| (Calico/Flannel)  |
+---------+---------+
          |
          v
+-------------------+
| Join Worker Nodes |
| kubeadm join      |
+---------+---------+
          |
          v
+-------------------+
| Verify Cluster    |
| kubectl get nodes |
+-------------------+
```

2. Kubernetes Components

API Server – The central management component that receives all requests and communicates with other cluster services.

etcd – A distributed key-value database used to store Kubernetes cluster configuration and state information.

Scheduler – Decides on which worker node a newly created Pod should run based on available resources and policies.

Controller Manager – Monitors cluster state and ensures the desired state is maintained automatically.

Kubelet – Runs on each worker node and manages Pods according to instructions received from the control plane.

Kube Proxy – Handles networking and load balancing between services and Pods.

Container Runtime – Software responsible for running containers, such as containerd or CRI-O.

3. Helm Chart

Helm is known as the package manager for Kubernetes. It simplifies application deployment by packaging all Kubernetes manifests into a reusable unit called a Helm Chart.

A Helm Chart typically contains: **Chart.yaml** – Metadata about the application. **values.yaml** – Customizable configuration values. **templates/** – Kubernetes resource templates. **charts/** – Dependent charts. Benefits of Helm: Easy deployment and upgrades. Version control for applications. Reusable templates. Simple rollback to previous versions. Consistent deployments across environments. Example: `helm install nginx bitnami/nginx`

This command downloads and deploys the Nginx application into a Kubernetes cluster using a predefined Helm chart.

Conclusion

Kubernetes provides a robust platform for container orchestration through its control plane and worker node components. Helm enhances Kubernetes by simplifying application deployment, upgrades, and configuration management, making it an essential tool for DevOps engineers.